

# Xuran Cai

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## Research Interests

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I am generally interested in Theoretical Computer Science (TCS), with a focus on graph theory and its algorithmic and structural aspects, which includes

- Program verification and formal methods
- SAT/SMT solving and constraint solving techniques
- Compiler optimization and program analysis
- Applications of theoretical methods to blockchain systems and distributed computation

## Education

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**Oxford-Max-Plank joint Program**, Doctor of Philosophy Sep 2025 - Present  
German & UK

- Joint Program by University of Oxford and Max Plank Institute
- Research Interest: Graph Theory, Verification, general TCS
- Supervisor: Prof. Amir Goharshady(Oxford) and Prof. Rupak Majumdar(MPI)

**Hong Kong University of Science and Technology**, Master of Philosophy Sept 2023 – June 2025  
Hong Kong SAR

- Thesis: Enhancing Compiler Optimization Efficiency through Grammatical Decompositions of Control-Flow Graphs
- Supervisor: Prof. Amir Goharshady and Prof. Jiasi Shen

**University of Wisconsin-Madison**, Bachelor of Science Sept 2019 – May 2023  
USA

- GPA: 3.86/4.0
- Major: Double Major in Computer Science and Mathematics
- Graduate with Distinction and Major Distinction in CS

## Publications

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**Extending Programmable Blockchain Protocols to Enable and Incentivize Parallel Execution of Smart Contracts** Sep 2026

T. Barakbayeva, X. *Cai*, Z. Cai, A.K. Goharshady

International Conference on Mathematical Research for Blockchain Economy, MARBLE'26

**Polynomial Invariant Generation for Floating Point Programs** July 2026

X. *Cai*, L. Chen, H. Fu

International Conference on Computer Aided Verification, CAV'26 (**CORE A\***, **CCF-A**)

**Series-Parallel-Loop Decompositions of Control-flow Graphs** February 2026

X. *Cai*, A.K. Goharshady, S. Hitarth, C.K. Lam

Journal of Systems Architecture, JSA'26

**Efficient Algorithms for Partial Constraint Satisfaction Problems over Control-flow Graphs** December 2025

X. *Cai*, A.K. Goharshady

International Symposium on Software Engineering: Theories, Tools, and Applications, SETTA'25

**Enhancing Compiler Optimization Efficiency through Grammatical Decompositions of Control-Flow Graphs** July 2025

X. *Cai*

Mphil Final Thesis

**Faster Chaitin-like Register Allocation via Grammatical Decompositions of Control-Flow Graphs**

March 2025

X. *Cai*, A.K. Goharshady, S. Hitarth, C.K. Lam

International Conference on Architectural Support for Programming Languages and Operating Systems, ASPLOS'25 (CORE A\*, CCF-A)

**Faster Lifetime-optimal Speculative Partial Redundancy Elimination for Goto-free Programs**

November 2024

X. *Cai*, A.K. Goharshady

International Symposium on Software Engineering: Theories, Tools, and Applications, SETTA'24

## Teaching Assistant Experiences

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**Design and Analysis of Algorithms**

HT Semester 2025-26

- Institute: Oxford
- Instructor: prof. Sandra Kiefer

**Functional Programming**

MT Semester 2025-26

- Institute: Oxford
- Instructor: prof. Andrzej Murawski

**Honors Discrete Mathematical Tools for Computer Science**

Fall Semester 2024-25

- Institute: HKUST
- Instructor: prof. Amir Goharshady

**Discrete Mathematical Tools for Computer Science**

Spring Semester 2023-24

- Institute: HKUST
- Instructor: prof. Sunil Arya and prof. Jiasi Shen

**Introduction to Computer Networks**

Spring Semester 2021-22

- Institute: UW-Madison
- Instructor: prof. Suman Banerjee

## Funding

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**Researcher Salary – MPI-SWS, Kaiserslautern, German**

Jan 2026 - present

**Department Scholarship – University of Oxford, Oxford, UK**

Oct 2025 - present

**Postgraduate Scholarship – HKUST, Hong Kong SAR**

Sep 2023 - July 2025

## Internship

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**Summer Intern Software Developer, Intel – Shanghai, China**

June 2021 - Aug 2021

- Worked on designing a service mesh structure with SGX hardware
- Designed and implemented the stereotype of the system

## Conference/Summer Schools

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**HUAWEI 2012 Lab Summer School** , Guangdong, China

Aug 17 - Aug 21, 2026

**CAV'26** , Lisbon, Portugal

July 26 - July 29, 2026

**Setta'25** , Oxford, UK

Dec 1 - Dec 3, 2025

**Setta'24** , Hong Kong SAR

Nov 26 - Nov 28, 2024

**FM'24** , Milan, Italy

Sep 9 - Sep 13, 2024

**Summer School on Discrete Mathematics**, Prague, Czech Republic

July 1- July 5, 2024